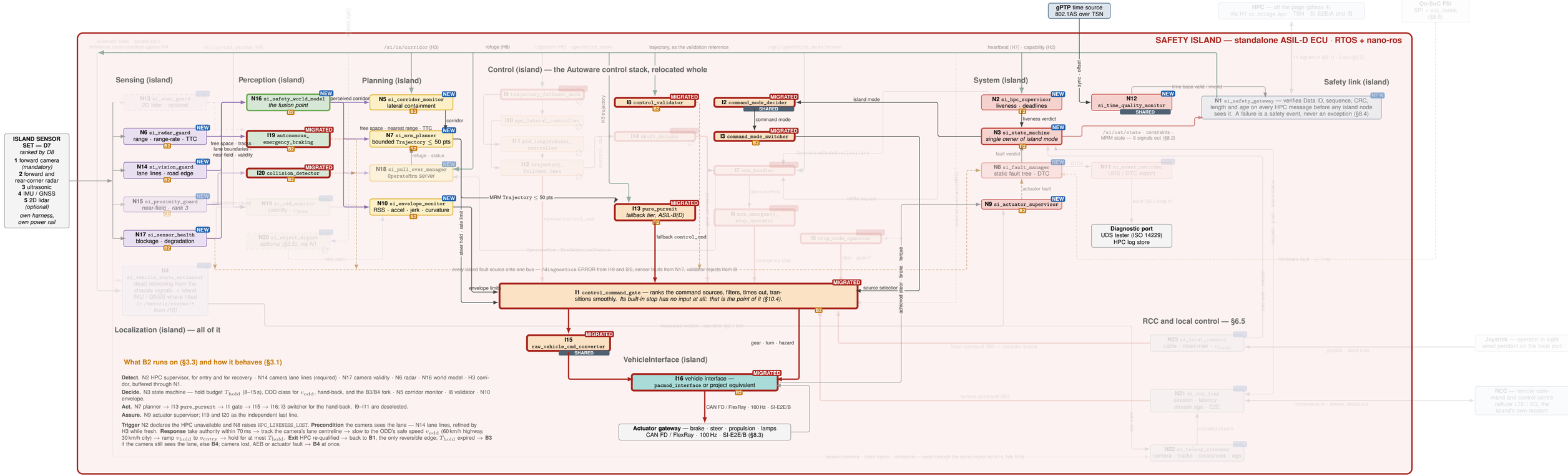
Phase 5 · B2 — Keep the vehicle in its lane after loss of the HPC

When the HPC is unavailable and B1 therefore cannot be used, take authority and hold the vehicle in its current lane — for a bounded 8–15 s window, first down to the ODD's safe speed — while the camera still sees the lane.
The phase-5 drawing, unchanged: what B2 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



What B2 runs on (§3.3) and how it behaves (§3.1)

Detect. N2 HPC supervisor, for entry and for recovery · N14 camera lane lines (required) · N17 camera validity · N6 radar · N16 world model · H3 corridor, buffered through N1.

Decide. N3 state machine — hold budget T_{hold} (8–15 s), ODD class for v_{odd}, hand-back, and the B3/B4 fork · N5 corridor monitor · I8 validator · N10 envelope.

Act. N7 planner → I13 `pure_pursuit` → I1 gate → I15 → I16; I3 switcher for the hand-back. I9–I11 are deselected.

Assure. N9 actuator supervisor; I19 and I20 as the independent last line.

Trigger N2 declares the HPC unavailable and N8 raises `HPC_LIVENESS_LOST`. **Precondition** the camera sees the lane — N14 lane lines, refined by H3 while fresh. **Response** take authority within 70 ms → track the camera's lane centreline → slow to the ODD's safe speed v_{odd} (60 km/h highway, 30 km/h city) → ramp v_{hold} to v_{entry} → hold for at most T_{hold}. **Exit** HPC re-qualified → back to **B1**, the only reversible edge; T_{hold} expired → **B3** if the camera still sees the lane, else **B4**; camera lost, AEB or actuator fault → **B4** at once.

Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED — the same Autoware node, migrated
NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour — B2
SHARED
B2
Used by no behaviour

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

HPC-sourced, verified by N1
island sensor data
control command into the gate
fault · `/diagnostics`
actuator feedback
state · mode · decision
optional (§3.5)

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

In this view — B2, lane-keep hold

full colour — named for B2 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B2's path by §3.1, but not named in §3.3

faded — not used by B2. An edge is drawn at the weaker of its two ends. Only B2's own tag is drawn on each box.